Ensemble Learning for Improved Multi-Class Brain Segmentation in Highly Imbalanced Datasets

***Abstract*—Multi-class segmentation of medical images presents many challenges due to the variety of imbalances present in the dataset combined with often small dataset sizes due to the costs and restrictions of acquiring data and creating detailed annotations. To improve the learning of rare and small classes, traditional techniques to address imbalance include data augmentation, balanced sampling schemes, and weighted loss functions which can all be used to emphasize learning of the minority classes. However, these techniques often struggle to maximally balance the performance of all classes, and in limited datasets may lead to early overfitting of small rare classes, yielding models with overall poor performance either of the small rare classes due to overfitting or of the rest of the classes which are undertrained due to early stopping. Furthermore, many techniques to prevent overfitting such as simplifying the model or introducing dropout may also affect the model's ability to learn extremely detailed small classes. This study presents an ensemble approach specialized to address the challenges of extreme class imbalance in a detailed brain lesion dataset. We demonstrate that by training separate models – one optimized for overall performance across majority classes and another specialized for the rare minority class - and then combining these models either using ensembling techniques or multi-stage learning, we can achieve improved segmentation results across all classes, including the rare class. Our results show a 22% improvement in performance using our imbalance focused combination model over our baseline model.**

I. Introduction

Medical image segmentation inherently is an imbalanced problem, with a majority background of blank space and non-segmented tissue, and foreground classes that come in all shapes, sizes, and prevalences. These imbalances are exacerbated even more in 3D medical imaging such as MRIs by adding another dimension going from area to volume imbalances. Furthermore, by the nature of the kinds of problems addressed by medical image segmentation, many classes of interest represent certain anatomical structures or pathologies which only exist in minority cases of diseases and which may present in a variety of ways, locations, shapes, sizes, and numbers. This variety of imbalances poses substantial difficulties in developing accurate and robust segmentation models for detailed annotations.

Conventional approaches to addressing class imbalance include sampling techniques, weighted loss functions, and extensive data augmentation. However, medical datasets often only represent a limited number of subjects and some of these methods may lead to overall suboptimal performance. Employing techniques like oversampling or a combination of over- and under-sampling to balance class representation under these conditions may cause the minority class to rapidly overfit. This creates a dilemma in the training process: to perform early stopping to maximize the performance on the minority class while undertraining other classes, or prioritize the performance of all classes resulting in a model that is completely overfit to or has otherwise failed to effectively and robustly learn the minority class. This occurs because the more prevalent classes dominate the loss landscape, leading to steeper descent and easier learning compared to the rare class. While data augmentation can help prevent overfitting and maximize the learning of especially minority classes in a limited dataset, data augmentation is complicated by the size and realistic limits of the dataset. For example, augmentation by rotation is limited for full brain scans if the model is set up for a registered brain orientation, and other transforms risk bringing the dataset outside the realm of realistic data. Generating new samples for minority classes on whole brain scans with a limited dataset size is arguably just as complicated of a problem or more so than the multi-class segmentation task itself.

This results in a fundamental challenge of learning all classes optimally by both addressing the inherent imbalances but also mitigating suboptimal conditions such as overfitting or catastrophic learning.

An alternative approach to this delicate balancing act is to train binary classifiers for each class separately. While this method may seem to circumvent some of the issues due to the class imbalance, it fails to capitalize on the rich inter-class information and feature developments that arise from multi-class learning.

To address these challenges, we propose a novel approach which involves training multiple models with different focuses:

- A generalist model optimized for overall performance across the majority of classes.
- A specialist model focused primarily on the performance of rare and minority classes.

By combining these models, we aim to capture both the robust features with inter-class information from the generalist model along with its overall good performance across classes, and the detail oriented features focused on boosting the rare minority class segmentation of the specialist model. This strategy allowed us to improve overall performance and near-match or improve the best segmentation performance for every class from individual models.

II. Data

A. Data Acquisition and Annotation

The dataset consists of 79 participants with chronic stroke-induced aphasia. All participants underwent neuroimaging at the FSM Center for Translational Imaging, with enrollment spanning three different sites. The dataset includes both T1 MPRAGE and FLAIR images for each participant.

Two experienced neuroradiologists performed manual segmentations using Freeview software. They annotated the co-registered T1 MPRAGE and FLAIR images, delineating seven distinct tissue categories:

1. Core infarct
2. Periinfarct gliosis
3. PVH
4. Right-sided LI
5. Left-sided LI
6. Right-sided WMH
7. Left-sided WMH

B. Data Processing

To ensure consistency and optimize the data for analysis, several preprocessing steps were implemented:

- Co-registration: For each subject, the T1, FLAIR, and segmentation mask images were co-registered to ensure spatial alignment.
- Resolution matching: The FLAIR images, originally acquired with 5 mm slice thickness, were upsampled to match the higher resolution of the T1 images (1 mm slice thickness).
- Bias Field Correction: To address intensity inhomogeneities in the MRI scans, which can affect segmentation accuracy, we applied N4 bias field correction. This step helps to mitigate the low-frequency, smooth intensity variations across the image that are caused by magnetic field inhomogeneities and variations in the sensitivity of the receiver coils. This preprocessing step is crucial for ensuring consistent intensity relationships between tissues across the entire image, which is particularly important for our multi-class segmentation task involving structures with subtle intensity differences.
- Brain extraction: A region of interest (ROI) mask encompassing the brain volume was generated using the ROBEX (Robust Brain Extraction) algorithm. This was used as a mask to focus on relevant brain tissues and exclude non-brain regions.
- Normalization: The data within the brain ROI was normalized to have zero mean and unit variance. This standardization step helps mitigate intensity variations across different scans and subjects, while not taking the non-brain portion of the scan into consideration.
- Class combination and extraction: To refine our classification targets, we performed two key modifications:
  - Two classes were relabeled for clarity. Core infarct became encephalomalacia, and periinfarct gliosis was shortened to just gliosis.
  - The left-sided and right-sided LI were combined into a single LI class. Similarly the left-sided and right-sided WMH was combined into just WMH. This consolidation was done to simplify the classification task and increase the sample size for these classes.
  - A new ventricles class was extracted using 3D Slicer software. This additional class was included to provide richer context for the periventricular hyperintensities class.

This brings the final tissue classes to:

1. Encephalomalacia
2. Ventricles
3. Gliosis
4. PVH
5. WMH
6. LI

C. Dataset Imbalance

The dataset exhibits significant class imbalance across the six annotated tissue categories, which poses challenges for model training and evaluation. As illustrated in Figure 1(a), the distribution of voxel counts shows the Encephalomalacia class is the most prevalent, with over 5 million voxels, while the LI class is the least common, with only around 30,000 voxels. This disparity results in the largest class being over 160 times more frequent than the smallest.

Furthermore, as shown in Figure 1(b), not all of the subjects have all of the annotated classes present, with only 29 subjects presenting LI. This is challenging for splitting the dataset as well as for sampling the data as effectively as possible during training.

D. Dataset Stratification

Given the unequal distribution of the six tissue classes across brain volumes, the dataset was split into 4 training folds and a testing set using a stratified approach. This stratification ensures that each fold as close as possible given the whole-brain subject constraint maintains representative proportions of each tissue category, crucial for addressing the inherent class imbalance in the data.

The stratification of data into folds, depicted in Figure 1(c) and Figure 1(d), maintains this imbalance as much as possible across training and validation folds and test set. Each fold contains a representative distribution of voxel counts and subject numbers per class, allowing for robust cross-validation while preserving the inherent dataset characteristics.

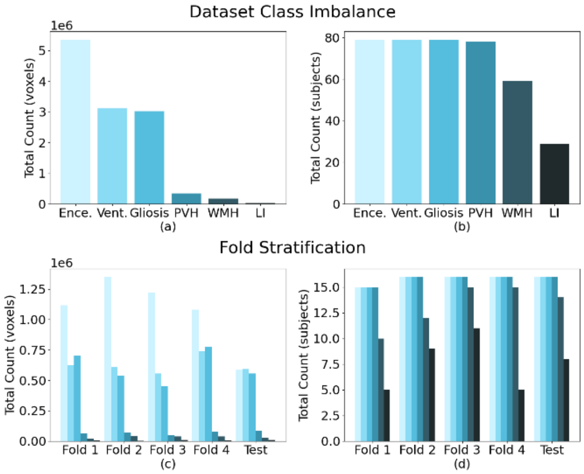

Fig. 1. *Visualization of dataset characteristics and fold stratification in the study. (a) Class imbalance by voxel count across six tissue categories, highlighting the distribution of voxel data. (b) Class imbalance by subject count, illustrating the number of subjects contributing to each tissue category. (c) Fold stratification showing the voxel distribution for each class within each fold. (d) Fold stratification showing the subject count contributing to each class.*

To achieve the desired ratio of classes and maintain the same number of subjects in each fold, we iteratively swapped subjects between folds while minimizing a loss function based on the relative error between the mean voxel sums per subject for each class and fold of the overall dataset and the current fold designations. The loss function was defined as for where V is the count of all voxels in class c and fold f, and for subjects S in fold f:

$$\bar{V}_{c,f} = \frac{V(c,f)}{S(f)} \tag{1}$$

$$\bar{V}_{c,all} = \frac{\sum_f V(c,f)}{\sum_f S(f)} \tag{2}$$

$$Error = \sum_f \sum_c \frac{|\bar{V}_{c,f} - \bar{V}_{c,all}|}{\bar{V}_{c,all}} \tag{3}$$

This approach resulted in a stratified dataset that maintained consistent class proportions across all folds and the same number of subjects.

III. METHODOLOGY

This section details the model architectures implemented for the full-brain generalist model and the patch-based specialist model as well as other techniques used to help address the dataset imbalance and learn a more robust segmentation model including soft labels, gradient accumulation, and weighted loss functions. These strategies were designed to enhance model robustness and ensure effective learning across all tissue categories.

A. *Full-Brain Model*

Our full-brain model utilizes a 3D U-Net architecture with specific modifications:

- Input Dimensions: We cropped the full brain images into 192x192x192 cube volumes. This standardization ensures aligned dimensions across all samples, facilitating consistent input data for the model. This size was chosen by finding the smallest bounding region of all of the brain tissue regions, and then choosing the smallest size divisible by a high power of 2, in this case 2^6, for ease of model development.
- Model Architecture: The 3D U-Net model was chosen for its proven effectiveness in medical image segmentation and ability to extract features from the 3D information.
- Training Strategy: The model was trained on stratified folds of the dataset, ensuring balanced representation of each class within training and validation sets.

B. *Full-Brain Model Data Augmentation*

To enhance the model's ability to generalize and learn rare classes, we implemented several data augmentation techniques. These augmentations aim to reduce overfitting and improve the robustness of the model by introducing variability in the training data.

- Random Flipping: The full brain images were randomly flipped along the sagittal axis, effectively swapping the left and right sides of the brain. This introduces symmetry variations, which help the model learn spatial invariance and improve its ability to generalize across different anatomical orientations.
- Voxel-wise Gaussian Noise: By adding random Gaussian noise to the voxel intensities, this method simulates acquisition variability and noise inherent in medical imaging. It enhances the model's robustness by training it to recognize features despite noise interference.
- Voxel-wise Scaling: This augmentation randomly scales voxel intensities, adjusting contrast and brightness. It aids in teaching the model to handle variations in imaging conditions.
- Brain-wise Scaling: Scaling entire brain volumes mimics different imaging conditions and patient sizes. This augmentation helps the model adapt to variations in anatomical size and shape across different subjects. The scaling was kept to within 10 degrees.
- Shear and Rotation: Applying shear transformations and rotations up to 10 degrees introduces slight geometric variations to simulate different anatomical orientations and help the model become invariant to small misalignments during data acquisition.

These augmentations were carefully chosen to maintain anatomical plausibility while increasing the diversity of training samples. This approach helps the model become more resilient to variations in input data and improves its ability to generalize to unseen cases.

C. *Enhancing Learning for Rare Classes*

To improve the model's ability to learn rare classes, we implemented several advanced techniques:

- Soft Labels: By using soft labels, we introduced uncertainty into the training process. This approach helps the model generalize better by allowing it to learn from a distribution of possible labels rather than hard, one-hot encoded targets. This can be particularly beneficial for classes that are physically smaller since the uncertainty near the boundary has a much larger affect on the performance metrics. We explored two different lable smoothing techniques using a 3D gaussian kernel. The first approach involved applying a 3D Gaussian kernal with a standard deviation of 1 to the original segmentation labels. This method creates a smoother transition at the boundary of the segmented regions while using a small sigma to preserve small lesions. The smoothed labels are generated by convolving each channel of the original segmentation mask with a 3D Gaussian kernel:

$$y_{\text{smooth}_c} = y_c * G \tag{4}$$

The second method, in addition to convolving the Gaussian kernel, also adds the original segmentation mask and clips the result to zero. This maintains a value of 1 inside the entirety of the ground truth segmentation while also adding benefits of smooth labeling in the near outside of the original segmentation.

$$y_{\text{smooth}_c} = min(y_c * G + y_c, 1) \tag{5}$$

- Gradient Accumulation: To effectively handle small batch sizes due to memory constraints, gradient accumulation was employed. This technique allows the model to accumulate gradients over multiple iterations before performing a weight update. It simulates a larger batch size, which can stabilize training and improve convergence, especially for underrepresented classes. This technique was intented to help specifically with the subject level imbalance in full-brain segmentation since for some batches there would not be any presence of the minority classes. However in testing, we did not find gradient accumulation to be notably beneficial.
- Weighted Loss Functions: We applied class weights in our loss function to address class imbalance directly. By assigning higher weights to rarer classes, the loss function emphasizes these classes more during training. This adjustment helps ensure that the model pays adequate attention to minority classes, improving their representation in the learned features.

D. *Patch-Based Model*

For our patch-based approach, we implemented a 3D U-Net model utilizing 32x32x32 voxel patches. This method allowed for more data augmentation and techniques to counteract imbalance without being limited to the full volume of the brain. For example it allowed the use of a sample scheme to perform a more balanced sample across all of the classes, and additional augmentations such as flipping in multiple dimensions or rotation by larger angles.

To address the extreme class imbalance in our dataset, we developed the following sampling scheme:

1. Class-Based Sampling: We randomly selected a class for each training sample.
2. Initial Patch Extraction: A larger 101x101x101 voxel patch was initially cropped around the selected voxel. This oversized patch served as the canvas for our augmentation techniques.
3. Augmentation in Larger Context: By performing augmentations on this larger patch, we ensured that the selected voxel (potentially from a minority class) remained at the center. This strategy was crucial for preserving the integrity of small lesions or rare tissue types during transformations.
4. Transformation Techniques: We applied the same set of augmentations as in our full-brain model, including random flipping, voxel-wise gaussian noise addition, voxel-wise gaussian scaling, shear transformations, and data scaling. Due to the nature of extracting patches, we flipped across all dimensions not just sagittal and allowed for full rotation in any spatial dimension.
5. Final Cropping: After augmentation, we cropped the transformed 101x101x101 patch down to the final 32x32x32 size. Importantly, the initially selected voxel was allowed to appear anywhere within this final crop, introducing further variability in the training data.

By employing this patch-based approach with our tailored sampling strategy, we aimed to improve the model's performance on underrepresented classes while maintaining its ability to capture local contextual information crucial for accurate segmentation.

To generate full-brain predictions using our patch-based model, we implemented a sliding window approach with several refinements. We employed a sliding window technique to process the entire brain volume using our 32x32x32 patch-based model. This approach allows us to apply our localized model across the full extent of the brain.

To optimize the balance between computational efficiency and prediction accuracy, we explored various stride lengths for the sliding window. Smaller Strides provided more overlapping predictions, potentially increasing accuracy but at a higher computational cost. Larger Strides reduced computation time but with potentially less refined predictions at patch boundaries. We conducted experiments to determine the optimal stride that balances accuracy and processing time.

Since our stride was typically less than our patch size, in overlapping regions we tested simple averaging of predictions as well as implementing a Gaussian weighting scheme to give a higher weight to the central voxels in overlapping patch predictions. This approach is based on the assumption that predictions for voxels at the center of a patch are likely more accurate than those at the edges due to having more contextual information during the training process.

E. *Ensembling Methods*

To leverage the strengths of both our generalist model (optimized for overall performance across majority classes) and our specialist model (focused on the minority class), we explored various ensembling techniques. These methods aim to combine the outputs of the two models effectively, balancing the overall segmentation performance with improved detection of the rare class. We investigated the following ensembling approaches:

- Simple Averaging: We directly averaged the output probabilities of both models for each voxel and class.
- Max Voting: For each voxel, we selected the class with the highest probability across both models.
- Z-score Normalization (Z-norm) Based Methods: We applied z-score normalization to both model outputs, then tested the average, max vote, and replacing the minority class predictions from the generalist model with those from the specialist model.
- Softmax-based Methods: we applied softmax to both model outputs, then tested the average, max vote, and replacing the minority class predictions from the generalist model with those from the specialist model.

The z-score normalization was applied to standardize the output distributions of both models before combining them, potentially mitigating any scaling differences between the models. Softmax was used to convert the raw model outputs into probability distributions, ensuring that the class probabilities for each voxel sum to one.

We evaluated each of these ensembling methods on our validation set to determine which approach provided the best balance between overall segmentation accuracy and improved performance on the minority class. The most effective method was then selected for our final model evaluation on the test set.

F. Two-Stage Model

To leverage the strengths of both our full-brain and patch-based approaches, we developed a two-stage model that combines their outputs for a final prediction. This approach aims to capture both global context and fine-grained local details, potentially improving overall segmentation accuracy.

In the first stage, we independently run both the full-brain and patch-based models. The Full-Brain Model processes the entire brain volume, capturing global contextual information and overall structural relationships. The Patch-Based Model analyzes localized 32x32x32 voxel regions, focusing on detailed features and potentially improving detection of small lesions or rare tissue types.

The second stage integrates the outputs from both models along with the original input data. We concatenate the original brain MRI data T1 and FLAIR modalities, the probability maps from the full-brain model, and the probability maps from the patch-based model.

A separate neural network processes this combined input. This model is designed to learn how to optimally weigh and integrate the information from both initial models and the original data.

In our second stage, we implemented a learnable weighting mechanism to dynamically adjust the importance of different input channels from our first-stage models. This approach allows the network to adaptively emphasize or de-emphasize features from the patch-based and full-brain models based on their relevance to the final segmentation task.

The learnable weights are implemented as a set of trainable parameters, one for each input channel. These weights are initialized to 1 and are updated during the training process through backpropagation.

G. Evaluation Metrics

To comprehensively assess our model's performance, we employed both standard volumetric metrics and specialized lesion-based metrics. This approach allows us to evaluate overall segmentation accuracy while also focusing on the model's ability to detect and delineate small lesions.

Our primary metric for evaluating segmentation performance is the Dice Similarity Coefficient (DSC). The DSC is widely used in medical image segmentation tasks due to its effectiveness in measuring spatial overlap between predicted and ground truth segmentations. It is defined as:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|} \tag{6}$$

where X and Y represent the predicted and ground truth segmentation masks, respectively.

To provide a more nuanced evaluation of our model's performance, particularly for small lesions, we adopted the lesion-based metrics proposed by Oguz et al.[9]. These metrics offer a detailed breakdown of various scenarios encountered in lesion detection and segmentation:

- Correct Detection: Instances where the model accurately identifies and segments a lesion.
- False Alarm: Cases where the model incorrectly predicts a lesion in a healthy region.
- Detection Failure: Situations where the model fails to detect an existing lesion.
- Merge: Occurrences where multiple distinct lesions are incorrectly segmented as a single lesion.
- Split: Cases where a single lesion is incorrectly segmented as multiple separate lesions.
- Split-Merge: Complex scenarios involving both splitting and merging of lesions in the same region.

These metrics provide valuable insights into the model's behavior, particularly in challenging cases involving small or closely spaced lesions. By analyzing these scenarios, we can better understand the strengths and limitations of our segmentation approach, especially in the context of rare or small tissue abnormalities.

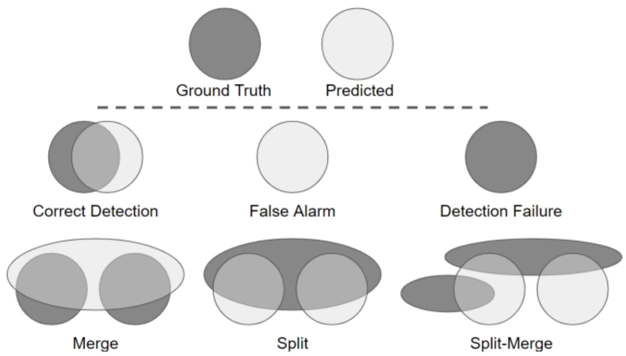

Fig. 2. *Visualization of the different scenarios encountered in lesion-based metrics. The ground truth is represented by the dark 'lesion' and the predicted segmentations are represented by the light 'lesion'. Under the dotted line visualizes the descriptive scenarious when the predicted segmentation does or does not align with the ground truth. Correct Detection: there is overlap between the predicted lesion and the ground truth lesion. False Alarm: a lesion was predicted but it has no overlap with the grount truth. Detection Failure: a ground truth lesion exists but no predicted segmentation overlaps with it. The bottom row describes slightly more complicated cases. Merge: multiple instances in the ground truth are incorrectly predicted as one lesion. Split: one instance in the grount truth is incorrectly predicted as multiple. Split-Merge: both splitting and merging of lesions occur in the same region.*

It is important to note that our dataset contained numerous single-voxel segmentations, likely due to noise in the labeling process. These single-voxel 'lesions' posed a challenge for the lesion-based metrics, as they may not represent clinically significant features. To address this, we either filtered out single-voxel lesions before applying these metrics or interpreted the results with caution, acknowledging that the metrics may not be 100% accurate in representing true lesion detection performance. This preprocessing step or interpretative approach was necessary to ensure that our evaluation focused on meaningful lesion detection and segmentation, rather than being skewed by labeling artifacts.

H. Loss Functions

In our study, we explored various loss functions to optimize our segmentation models, each serving different purposes in addressing the challenges of a dataset with an extreme imbalance and small lesions.

- Dice Loss is a widely used loss function in medical image segmentation due to its effectiveness in handling class imbalance. It is based on the Dice coefficient which measures the overlap between the predicted segmentation and the ground truth and is our main metric for gauging segmentation performance. Multi-class Dice loss is defined as:

$$L_{Dice} = 1 - \frac{1}{C}\sum_{c=1}^{C} \frac{2\sum_i^N \hat{p}_{ci} y_{ci}}{\sum_i^N \hat{p}_{ci}^2 + \sum_i^N y_{ci}^2} \tag{7}$$

where C is the number of classes, N is the total number of voxels, $\hat{p}_{ci}$ is the predicted probability for class c at voxel i, and y_{ci} is the ground truth label for class c at voxel i. Dice loss decreases to zero as the prediction approaches perfect overlap with the ground truth. Because it focuses on the overlap of positive regions, it is less sensitive to the majority background class in medical image segmentation.

- Cross-Entropy (CE) Loss is another widely used loss function for image segmentation tasks that measures the dissimilarity between the predicted probability distribution and the true distribution of the classes. Multi-class Cross Entropy Loss is defined as:

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ci} \log \hat{p}_{ci} \tag{8}$$

where C is the number of classes, N is the total number of voxels, $\hat{p}_{ci}$ is the predicted probability for class c at voxel i, and y_{ci} is the ground truth label for class c at voxel i.

- Combined Dice and Cross Entropy Loss (DiceCE): We combined Dice and Cross-Entropy losses to leverage the strengths of both and help learn more unique features:

$$L_{DiceCE} = \alpha\, L_{Dice} + (1-\alpha)\, L_{CE} \tag{9}$$

- Combined Dice and Cross Entropy Loss with class weights: To address the class imbalance and emphasize certain classes, we implemented a combination of cross-entropy and Dice loss, incorporating class-specific weights:

$$L_{Dice}(w) = 1 - \frac{2\sum_{c=1}^{C} w_c \sum_i^N \hat{p}_{ci} y_{ci}}{\sum_{c=1}^{C} w_c \left(\sum_i^N \hat{p}_{ci}^2 + \sum_i^N y_{ci}^2\right)} \tag{10}$$

$$L_{CE}(w) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} w_c y_{ci} \log \hat{p}_{ci} \tag{11}$$

$$L_{DiceCE}(w) = \alpha\, L_{Dice}(w) + (1-\alpha)\, L_{CE}(w) \tag{12}$$

where α is the balancing factor between Dice loss and CE loss, w_c is the weight for class c, C is the number of classes, N is the total number of voxels, $\hat{p}_{ci}$ is the predicted probability for class c at voxel i, and y_{ci} is the ground truth label for class c at voxel i.

This combined weighted loss function allows us to balance between cross-entropy and Dice loss, apply class specific weights to address the class imbalance, and emphasize certain classes during training for 2 stage models.

- Gaussian-Masked Loss for Patch Segmentation: For patch-based segmentation, we implemented a Gaussian-masked loss to prioritize the center of the prediction:

$$Gaussian - Masked\ Loss = Loss \cdot G(\sigma) \tag{13}$$

where G is a 3D Gaussian function centered at the patch center with standard deviation σ.

- Focal Loss: To address class imbalance, we explored Focal Loss, which down-weights well-classified examples:

$$L_{Focal} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ci}(1-\hat{p}_{ci})^{\gamma} \log(\hat{p}_{ci}) \tag{14}$$

where N is the total number of voxels, C is the number of classes, $\hat{p}_{ci}$ is the predicted probability for class c at voxel i, y_{ci} is the ground truth label for class c at voxel i, and γ is the focusing parameter.

- Boundary Loss: Boundary loss is a useful loss function for imbalance especially multi-dimensional imbalance, since it focuses on the contours of the segmented regions and penalizes discrepancies between the predicted and ground truth boundaries. This reduces the effective dimension of imbalance from 3D to 2D in our case. However the effectiveness of boundary loss is limited on our dataset due to the noisy labeling and extremely small lesion sizes.

$$L_{Boundary} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}\left|\phi_{\hat{p}_{ci}} - \phi_{y_{ci}}\right| \quad (15)$$

where $\phi_{\hat{p}_{ci}}$ is the distance transform of the predicted segmentation for class c at voxel i, and $\phi_{y_{ci}}$ is the distance transform of the ground truth segmentation for class c at voxel i.

- F-beta Loss: To address the challenge of segmenting small lesion classes and prioritize sensitivity in our model, we employed the F-beta score as a component of our loss function. The F-beta score is a weighted harmonic mean of precision and sensitivity, allowing us to adjust the balance between these two metrics to, in our case, emphasize the sensitivity to prioritize finding small lesions.

$$Sensitivity\ (recall) = \frac{\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ci}\hat{p}_{ci}}{\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ci}} \quad (16)$$

$$Precision = \frac{\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ci}\hat{p}_{ci}}{\sum_{i=1}^{N}\sum_{c=1}^{C} \hat{p}_{ci}} \quad (17)$$

$$F_{\beta} = (1+\beta^2)\frac{precision \cdot recall}{(\beta^2 precision) + recall} \quad (18)$$

$$L_{F_{\beta}} = 1 - F_{\beta} \quad (19)$$

where β is a positive real number. When $\beta = 1$ the F_{β} score reduces to the Dice coefficient or F_1 score. When $\beta > 1$, more weight is given to the sensitivity portion of the calculation, emphasizing locating small lesions in our case and deemphasizing precise segmentation.

IV. Results and Discussion

A. Overall Results

Figure 3 shows the full-brain, patch-based, and ensemble segmentation results versus the baseline model without any augmentation or imbalance methods. The LI class undergoes complete learning failure due to overfitting, and all of the other classes underperform compared to our other models.

	DSC						
	Enceph.	Vent.	Gliosis	PVH	WMH	LI	Average
Full-Brain Baseline	0.6462	0.5805	0.6604	0.3124	0.1553	0.0000	0.3925
Full-Brain + Imbalance Methods	0.6995	**0.6103**	0.7027	**0.3581**	0.2960	0.0561	0.4538
Patch-Based	0.2729	0.3696	0.5513	0.3254	0.1860	0.1720	0.3129
Ensemble	**0.7050**	0.6085	**0.7116**	0.3536	**0.3159**	**0.1730**	**0.4779**

Fig. 3. *Comparison of Full-Brain, Patch-Based, and Ensemble Model with Baseline Full-brain model. This table shows the segmentation performance of the best performing models with imbalance combatting techniques compared to the baseline model with no technique implemented.*

B. Patch-Based Results

Figures 4 and 5 show patch-based model results comparing different stride length and sigma values for an applied Gaussian mask during inference.

		DSC						
stride	sigma	Enceph.	Vent.	Gliosis	PVH	WMH	LI	Average
8	16	**0.2808**	**0.4374**	**0.5530**	**0.3264**	0.1851	0.1365	**0.3199**
16	16	0.2725	0.3703	0.5514	0.3218	**0.1872**	**0.1710**	0.3124
32	16	0.2528	0.2763	0.5135	0.3055	0.1633	0.1110	0.2704

Fig. 4. *Comparison of Patch-Based Models with Varying Stride Lengths: This table shows the segmentation performance of patch-based models using different stride lengths during inference. The models were evaluated on the test set, with dice score results reported for each of the six foreground classes (Encephelomalacia, Ventricles, Gliosis, PVH, WMH, and LI) as well as the average Dice score. Stride lengths of 8, 16, and 32 were tested to assess trade-off between computational efficiency and segmentation performance.*

		DSC						
stride	sigma	Enceph.	Vent.	Gliosis	PVH	WMH	LI	Average
16	1	0.2642	0.3375	0.5401	0.3204	0.1754	0.1545	0.2987
16	5	0.2648	0.3410	0.5409	0.3244	0.1786	0.1647	0.3024
16	10	**0.2729**	0.3696	0.5513	**0.3254**	0.1860	**0.1720**	**0.3129**
16	16	0.2725	**0.3703**	**0.5514**	0.3218	**0.1872**	0.1710	0.3124
16	32	0.2709	0.3676	0.5498	0.3187	0.1855	0.1712	0.3106
16	n/a	0.2702	0.3662	0.5489	0.3174	0.1849	0.1704	0.3097

Fig. 5. *Comparison of Patch-Based Models with Varying Sigma Value: This table shows the segmentation performance of patch-based models using Gaussian masks with different sigma values during inference. The n/a entry performed a simple average without any Gaussian mask.*

C. Ensemble Results

The performance of various ensemble methods compared to the original models is presented in Fig. 6. This comprehensive comparison reveals the effectiveness of our ensemble approach in addressing the challenges posed by the extreme class imbalance in our multi-class brain segmentation task.

Among the ensemble methods tested, the softmax average technique emerged as the top performer. This method not only outperformed all other models in segmenting the LI (Lesion Infarct) and WMH classes which were our primary focus due to their and the two smallest classes, but also maintained superior or near-superior performance across all other classes.

The softmax average ensemble demonstrated a remarkable ability to leverage the strengths of both the generalist and specialist models. This approach proved particularly beneficial for the LI class, where the ensemble outperformed all other models. Notably, the performance improvement in the LI class did not come at the expense of accuracy in other classes.

The results highlight the potential of carefully designed ensemble methods to overcome the limitations of individual models in highly imbalanced multi-class segmentation tasks.

	DSC						
	Enceph.	Vent.	Gliosis	PVH	WMH	LI	Average
Models							
Fullbrain Only	0.6995	**0.6103**	0.7027	0.3581	0.2960	0.0561	0.4538
Patches Only	0.2729	0.3696	0.5513	0.3254	0.1860	0.1720	0.3129
Ensemble Methods							
Average	0.7080	0.6084	**0.7152**	0.3596	0.3020	0.0786	0.4620
Max Vote	0.6996	**0.6103**	0.7029	0.3589	0.2965	0.0761	0.4574
Softmax Average	0.7050	0.6085	0.7116	0.3536	**0.3159**	**0.1730**	**0.4779**
Softmax Max Vote	0.7016	0.6083	0.7121	**0.3603**	0.2967	0.0847	0.4606
Softmax LI Replace	0.6995	**0.6103**	0.7028	0.3577	0.2980	0.0850	0.4589
Z-Norm Average	**0.7081**	0.6020	0.7105	0.3545	0.3094	0.1136	0.4664
Z-Norm Max Vote	0.7080	0.6046	0.7134	0.3557	0.3013	0.0730	0.4593
Z-Norm LI Replace	0.6995	**0.6103**	0.7027	0.3581	0.2980	0.0679	0.4561

Fig. 6. *Table of dice score results for different ensemble methods compared to the original models. The best performing ensemble model, softmax average,*

outperforms all models on LI segmentation, and maintains the best result or near best result in all other classes.

D. Computational Efficiency

In our study, we observed significant differences in computational requirements across our various methodologies, each presenting unique trade-offs between performance and efficiency.

The full-brain approach, while providing a comprehensive view of the entire brain volume, demanded substantial computational resources. It required larger memory allocation due to processing entire brain volumes simultaneously and necessitated smaller batch sizes to fit within GPU memory constraints, potentially affecting training stability and convergence speed. However, it benefited from faster inference times once trained, as entire volumes could be processed in a single forward pass.

Our patch-based approach, while more memory-efficient, introduced its own set of computational challenges. It allowed for larger batch sizes due to smaller input dimensions, potentially improving training stability. However, it significantly increased training time due to the need to process multiple patches per brain volume. The complex sampling scheme and extensive data augmentations, while crucial for addressing class imbalance and improving generalization, were computationally expensive and prolonged the training process. Inference time was notably longer compared to the full-brain method, as it required aggregating predictions from multiple patches to reconstruct full brain segmentations.

The implementation of sophisticated loss functions added another layer of computational complexity. Focal Loss increased computational overhead due to its element-wise operations and power calculations. Boundary Loss required additional computations for distance transform calculations, further increasing the computational load.

Overall, the full-brain method offered faster inference but was limited by GPU memory constraints during training. The patch-based approach allowed for more extensive data augmentation and finer control over sampling but at the cost of increased training and inference times. Advanced loss functions provided unique feature learning but added to the overall computational burden.

Future work could explore optimizations such as mixed-precision training or more efficient data augmentation techniques to mitigate some of these computational challenges.

E. Conclusion

In conclusion, this study demonstrates the effectiveness of an ensemble approach in addressing the challenges of extreme class imbalance in multi-class brain segmentation tasks. By combining a generalist model optimized for overall performance with a specialist model focused on the rarest class, we achieved improved segmentation results across all classes, including notable enhancements in rare class detection. The softmax average ensemble method proved particularly effective, outperforming individual models and maintaining superior or near-superior performance across all classes. This approach not only improved the segmentation of the challenging LI class but also preserved high accuracy for more prevalent classes. Our findings highlight the potential of carefully designed ensemble methods to overcome the limitations of individual models in highly imbalanced multi-class segmentation tasks. This research contributes to the field of medical image analysis by offering a robust strategy for handling detailed, imbalanced datasets, potentially leading to more accurate and reliable diagnostic tools in clinical practice. Future work could explore the application of this ensemble approach to other complex medical imaging tasks and investigate its potential for improving patient outcomes through enhanced diagnostic accuracy.

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